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The Positive and Negative Affect Schedule (PANAS): Construct validity, measurement properties and normative data in a large non-clinical sample

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Objectives. To evaluate the reliability and validity of the PANAS (Watson, Clark, & Tellegen, 1988b) and provide normative data.

Design. Cross-sectional and correlational.

Method. The PANAS was administered to a non-clinical sample, broadly representative of the general adult UK population (N = 1,003). Competing models of the latent structure of the PANAS were evaluated using confirmatory factor analysis. Regression and correlational analysis were used to determine the influence of demographic variables on PANAS scores as well as the relationship between the PANAS with measures of depression and anxiety (the HADS and the DASS).

Results. The best-fitting model (robust comparative fit index = .94) of the latent structure of the PANAS consisted of two correlated factors corresponding to the PA and NA scales, and permitted correlated error between items drawn from the same mood subcategories (Zevon & Tellegen, 1982). Demographic variables had only very modest influences on PANAS scores and the PANAS exhibited measurement invariance across demographic subgroups. The reliability of the PANAS was high, and the pattern of relationships between the PANAS and the DASS and HADS were consistent with tripartite theory.

Conclusion. The PANAS is a reliable and valid measure of the constructs it was intended to assess, although the hypothesis of complete independence between PA and NA must be rejected. The utility of this measure is enhanced by the provision of large-scale normative data.

The Positive and Negative Affect Schedule (PANAS) is a 20-item self-report measure of positive and negative affect developed by Watson, Clark, and Tellegen (1988b). NA and

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PA reflect dispositional dimensions, with high-NA epitomized by subjective distress and unpleasurable engagement, and low NA by the absence of these feelings. By contrast, PA represents the extent to which an individual experiences pleasurable engagement with the environment. Thus, emotions such as enthusiasm and alertness are indicative of high PA, whilst lethargy and sadness characterize low PA (Watson & Clark, 1984). It has, however, been argued that the labels *positive affect* and *negative affect* are misleading. Watson, Wiese, Vaidya, and Tellegen (1999) point out that PA and NA are predominantly defined by the *activation* of positively and negatively valenced affects, respectively (i.e. the lower ends of each dimension are typified by its absence). Thus, to emphasize the activated nature of each of these constructs, it has been argued that positive affect and negative affect should be renamed positive activation and negative activation, respectively. In the present work, these labels are to be regarded as interchangeable.

The PANAS is claimed to provide *independent* measures of PA and NA. Since its development the measure has been employed in research for diverse purposes. Its popularity may be attributed to its brevity and, perhaps more important, its close association with an influential conceptualization of anxiety and depression: the tripartite model (Clark & Watson, 1991b).

Although anxiety and depression are phenomenologically distinct, it has proven very difficult to distinguish between these constructs by empirical means, either through the use of clinicians' ratings or, particularly, through the use of self-report measures (Clark & Watson, 1991a). It has been suggested that this is because most existing self-report scales for anxiety and depression predominantly measure the common factor of negative affectivity (Watson & Clark, 1984). There is a great deal of evidence in support of this position (Cole, 1987; Feldman, 1993). The tripartite model, however, posits that in addition to the *common* factor of negative affectivity there are *specific* components to anxiety and depression that allow them to be differentiated. In the case of anxiety, the specific component is physiological hyperarousal and in the case of depression, it is low PA (low PA is similar to the psychiatric concept of anhedonia).

Watson *et al.* (1995b) directly tested the predictions of the tripartite model. Three student, one adult and one patient sample were administered the Mood and Anxiety Symptom Questionnaire (MASQ; Watson & Clark, 1991). The MASQ consists of three 'general' depression and anxiety scales as well as two specific scales, one of which measures anxious arousal, the other anhedonic depression. Across samples, the pattern of correlations consistently revealed that the specific measures possessed superior discriminant validity in comparison with the general depression and anxiety measures. Moreover, this was achieved without compromising convergent validity. Using the same five samples, Watson *et al.* (1995a) conducted an exploratory factor analysis (EFA) of the MASQ. A three-factor solution corresponding to general distress, anhedonic depression and somatic anxiety could be extracted for all five groups. Thus, both studies provide preliminary support for the validity of the tripartite model.

In addition, the PANAS has been shown to be effective at differentiating between depression and anxiety in clinical samples. Dyck, Jolly, and Kramer (1994) conducted an EFA of self-report measures of anxiety, depression, NA and PA in a psychiatric sample (N = 162) that identified two correlated factors ($r = -.32$) corresponding to NA and PA. Regression analyses were used to test whether these factors predicted self-reported anxiety and depression in independent clinical measures. The NA factor, and not the PA factor, significantly contributed to predicting anxiety, but *both* factors significantly predicted depression. Analogously, Jolly, Dyck, Kramer, and Wherry

(1994) administered a diverse range of clinical measures to 159 psychiatric out-patients. Partial correlations revealed that when NA was controlled, depression, but not anxiety, was related to PA scores. However, with PA controlled, both depression and anxiety explained a substantial amount of the variance in NA scores. Both these studies support tripartite theory, as they found PA to be specifically related to depression and not anxiety, but NA to be highly related to both.

If the use of the PANAS in research, and potentially in clinical practice, is to be optimal then it is necessary to delineate its underlying structure. Watson *et al.* (1988b) conducted an EFA with varimax rotation that revealed that the first two factors accounted for a very high proportion of the common variance, with all items loading cleanly on their designated factor. However, confirmatory factor analysis (CFA) provides an alternative to EFA and is associated with a number of advantages. In particular, in CFA, the fit of competing models can be compared quantitatively.

To date, two studies involving the 20-item PANAS have employed CFA. Crocker (1997) reported that, although globally Watson *et al.*'s (1988b) oblique two-factor model represented a good fit to data derived from a youth sporting group ($N = 671$), a degree of misspecification remained. Crocker (1997) suggested that this was because the items comprising the scale were derived from various content areas (Zevon & Tellegen, 1982), and therefore permitting correlated error would improve the model's fit. However, no study to date has tested a model parameterized according to these specifications. Moreover, Mehrabian (1998) found that a complex hierarchical structure represented a superior, though still inadequate fit, in comparison to Watson *et al.*'s (1988b) hypothesized two-factor structure. However, this model was derived on a purely *a posteriori* basis and thus may have arisen as a consequence of an idiosyncratic sample, or may simply have reflected the fact that the complex model was less restricted. Again, no study to date has attempted to determine which of these possibilities is correct.

The most controversial characteristic of the PANAS is the purported independence of its subscales. It has been argued that it is counter-intuitive to regard happiness and sadness as unrelated constructs (Costa & McCrae, 1980) and indeed all measures of PA and NA developed prior to the PANAS have proven at least moderately negatively correlated. Watson *et al.* (1988b) argue that this is attributable to inadequacies in the instruments themselves and not the notion of independence. Watson *et al.* (1988b) have reported low to moderate correlations between the PA and NA scales, ranging from $-.12$ to $-.23$, with other studies reporting similar results (Chen, Dai, Spector, & Jex, 1997; Joiner & Blalock, 1995; Mehrabian, 1998). Moreover, using CFA, two nearly orthogonal dimensions of PA and NA were reported for a 10-item short form of the PANAS ($r = -.10$; Mackinnon *et al.*, 1999).

However, Green, Goldman, and Salovey (1993) argue that random measurement error and acquiescence attenuates negative correlations, and thus that two dimensions such as PA and NA may appear to be relatively independent when in fact they are opposite poles of the same dimension. Moreover, van Schuur and Kiers (1994) argue that an artifact of factor analysis when analysing bipolar concepts is the identification of two factors, as the two halves of the same dimension are treated as independent. Thus, alternative models that reflect bipolarity may account for the nearly independent dimensions found in factor analysis. Indeed, Russell and Carroll (1999) maintain that bipolarity represents the most parsimonious fit to models of PA and NA, and that previous research has erroneously assumed that a necessary consequence of bipolarity is an invariant latent correlation between the two constructs. In fact, the correlation

changes as a function of time-frame, response format, and as exemplified by the PANAS, the items comprising the instrument.

Russell and Carroll's (1999) semantic analysis of affect terms delineates two dimensions corresponding to pleasantness and activation. The relationship between PA and NA is argued to be dependent upon which item clusters are chosen as, whilst some are semantically opposite, others are independent. Inspection of the items comprising the PANAS suggests that relatively independent subscales have been attained through omitting affective terms which represent low activation, i.e. focusing on a restricted range of affect. Whilst the NA items represent the cluster 'highly unpleasant/high activation', the PA items represent the cluster 'highly pleasant/high activation', these two clusters are 90° apart on Russell and Carroll's (1999) model. This raises questions regarding the construct validity of the PANAS scales as, if selection of items was explicitly guided by a quest for orthogonality, the items may not adequately represent their putative constructs. This possibility is supported by the fact that Watson and Clark (1984) state that fear is 'entirely unrelated to NA' (p. 469), yet Watson *et al.* (1988b) include both 'scared' and 'afraid' in the NA scale.

However, although questions have been raised regarding whether it is appropriate to regard the constructs of PA and NA as relatively independent, there is a great deal of evidence that they are distinct (see Watson *et al.*, 1999). In particular, Watson *et al.* (1999) argue that the dimensions of PA and NA represent the subjective components of the more general biobehavioural systems of approach and withdrawal, or the behavioural inhibition system (BIS) and behavioural engagement system (BES), respectively. It is suggested that, since the purpose of the BIS is to inhibit behaviour that may lead to undesirable consequences, the negative emotional states that characterize the NA dimension help to promote vigilant apprehensiveness. Relatedly, the purpose of the BES system is to ensure that essential resources are obtained, and thus the positive emotional states associated with PA can be regarded as motivating goal-directed behaviours. Watson *et al.* (1999) argue that whilst these adaptive systems are separate, they are not entirely independent of one another, thus accounting for the moderate correlations reported between the constructs of PA and NA.

Tellegen, Watson, and Clark (1999) suggest that an overarching bipolar happiness-versus unhappiness dimension underlies the negative correlation between NA and PA. Tellegen *et al.* (1999) found, using exploratory hierarchical factor analysis, that the structure of affect consists of a three-level hierarchy, in which the most general dimension of happiness–sadness forms a largely bipolar structure, at the second level PA and NA are relatively independent, and at the lowest level there are more circumscribed discrete emotions. Whereas, when using CFA the latent correlation between happiness–sadness with high ($r = -.91$), the correlation between PA and NA was substantially smaller ($r = -.43$). In the exploratory analyses it was also found that PA and NA were only moderately correlated ($r = -.31$).

As stated previously, the PANAS has been extensively employed, and this is reflected in the fact that shortened, elongated, and children's versions have been developed. It is therefore surprising that there have been relatively few studies of other aspects of the instrument's psychometric properties. Watson *et al.* (1988b) administered the PANAS with time-frames ranging from 'right now' to 'during the last year' to a large, predominantly student, sample. The reliability of the PA scale ranged from .86 to .90, the NA scale from .84 to .87; values similar to those obtained from independent research involving clinical and non-clinical populations (Jolly *et al.*, 1994; Mehrabian, 1998; Roesch, 1998). However, the non-clinical studies that have typically been

conducted either employed purely student samples (Roesch, 1998), or participants not broadly representative of the general population (Mehrabian, 1998; Watson *et al.*, 1988b). The nature of these samples means that the generalizability of their results to the normal population is uncertain (Gotlib, 1984; Nezu, Nezu, & Nezu, 1986).

Normative data are also very limited; a literature search conducted by the present authors failed to uncover general adult population norms for the English-language version derived from a large sample. Moreover, despite the fact that three studies involving clinical populations have utilized the 'past week' time format (Dyck *et al.*, 1994; Jolly *et al.*, 1994; Kuiper, McKee, Kazarian, & Olinger, 2000), no study conducted in the general population has done so. Relatedly, the influence of demographic characteristics on PANAS scores has also gone largely uninvestigated.

The aims of the present study were (1) to evaluate competing models of the latent structure of the PANAS using confirmatory factor analysis. Details of the parameterization of the models (and the theoretical, methodological and empirical considerations that guided their selection) are presented in the Methods section. The second aim (2) was to test whether the relationships of PA and NA with measures of depression and anxiety support tripartite theory. Thus (2a), it was hypothesized that the correlation between PA and depression would be significantly higher than the correlation between PA and anxiety. In addition (2b), it was hypothesized that PA would explain a significantly greater proportion of the variance unique to depression than would NA. Finally, we aimed to (3) estimate the reliability of the PANAS, (4) investigate the influence of demographic variables on PANAS scores, and (5) provide normative data for the PANAS.

Method

Participants

Complete PANAS data were collected from 1,003 members of the general adult population (females = 537, males = 466). Participants were recruited from a wide variety of sources including commercial and public service organizations, community centres, and recreational clubs. The mean age of the sample was 42.9 years (*SD* = 15.7) with a range of 18–91 years. The mean number of years of education was 13.7 (*SD* = 3.4).

Each participant's occupation was coded using the Office of Population Censuses and Surveys (1990) classification of occupations. Retired participants, participants who were currently unemployed, and those describing themselves as househusbands/housewives were coded by their previous occupations. Those who had never worked were coded as 5 (i.e. unskilled).

The percentage of participants in the occupational categories of professional (1), intermediate (2), skilled (3), semi-skilled (4) and unskilled (5) was 12, 42, 30, 6 and 10% respectively. The corresponding percentage for each category in the general adult population census is 7, 32, 42, 14 and 5%, respectively. Thus, whilst there is a broad range of occupational backgrounds in the present sample, there is a slight over-representation of professional occupations, and a slight under-representation of skilled and semi-skilled occupations. The percentage of participants in each of four age bands (18–29, 30–44, 45–59, 60+) was 23, 33, 28 and 16%. The corresponding percentage for each age band in the general adult population census is 27, 25, 22 and 26% respectively.

Again it can be seen that there is a broad spread, although there is a relative under-representation of individuals in the oldest age group.

Materials

Each potential participant received an introductory letter, a PANAS form, and a form for recording demographic variables. In addition, the majority of participants also received and completed self-report measures of depression and anxiety; the Depression Anxiety and Stress Scales (DASS; $N = 740$) and the Hospital Anxiety and Depression Scale (HADS; $N = 989$). Neither of these subgroups differed significantly from the overall sample with respect to age or gender. Participants sealed the completed forms in envelopes that were either collected by the researcher or returned by mail. The refusal rate was approximately 19%.

The Positive and Negative Affect Schedule (PANAS)

The PANAS (Watson *et al.*, 1988b) consists of two 10-item mood scales and was developed to provide brief measures of PA and NA. The items were derived from a principal components analysis of Zevon and Tellegen's (1982) mood checklist; it was argued that this checklist broadly tapped the affective lexicon. Respondents are asked to rate the extent to which they have experienced each particular emotion within a specified time period, with reference to a 5-point scale. The scale point are: 1 '*very slightly or not at all*', 2 '*a little*', 3 '*moderately*', 4 '*quite a bit*' and 5 '*very much*'. A number of different time-frames have been used with the PANAS, but in the current study the time-frame adopted was 'during the past week'.

The Depression Anxiety and Stress Scales (DASS)

The DASS (Lovibond & Lovibond, 1995) consists of three 14-item self-report scales that measure depression, anxiety and stress. The DASS was developed to quantify these disorders in both normal and clinical populations. Items are rated on a 4-point scale using a time-frame of 'over the past week'.

The Hospital Anxiety and Depression Scale (HADS)

The HADS was developed by Zigmond and Snaith (1983) to provide a brief means of identifying and measuring severity of depression and anxiety in non-psychiatric clinical environments. It consists of 14 items, seven of which measure depression, and the other seven anxiety. The items comprising the depression scale are predominantly based on the anhedonic state, so should be particularly related to PA.

Statistical analysis

Basic statistical analyses were conducted using SPSS Version 8. Confidence limits on Cronbach's α were derived from Feldt's (1965) formula. CFA (robust maximum likelihood) was performed on the variance–covariance matrix of the PANAS items using EQS for Windows Version 5.4 (Bentler, 1995). The fit of CFA models was assessed using the Satorra-Bentler scaled χ^2 statistic (S-Bχ^2), the robust comparative fit index (RCFI), the standardized root mean squared residual (SRMR) and the root mean squared error of approximation (RMSEA; Steiger, 2000). Hu and Bentler (1998, 1999) demonstrated, using Monte Carlo analyses, that the combination of the SRMR and RMSEA minimizes the rejection of well fitting models, yet possesses optimal sensitivity to model

misspecification. For the SRMR a cut-off value close to .08 or below is recommended, whilst for the RMSEA a cut-off of < .06 is recommended (Hu & Bentler, 1999).

It is possible to directly test whether more constrained models have significantly poorer fit than less constrained models; this feature of CFA is one of its major advantages over EFA. In the present case there is a slight complication because the Satorra-Bentler χ^2 statistic (S-Bχ^2) is used as an index of fit rather than the standard χ^2 statistic (the Satorra-Bentler statistic is recommended when the raw data are skewed). The *difference* between S-Bχ^2 for nested models is typically not distributed as χ^2 (Satorra, 2000). However, Satorra and Bentler (2001) recently developed a scaled-difference χ^2 test statistic that can be used to compare S-Bχ^2 from nested models. This statistic is used in the present study.[1]

Parameterization of competing models of the PANAS

The first model (Model 1a) to be evaluated was a single-factor model; this model expressed the hypothesis that the variance in the PANAS can be partitioned into one general factor plus error variance associated with each individual item (error variance here refers to the combination of true variance in the item that is independent of the factor plus random error). It is standard practice to test the fit of a one-factor model because it is the most parsimonious of all possible models. However, in the case of the PANAS, this model can also be seen as an expression of an intuitive hypothesis that the PANAS items measure opposite ends of a *single* dimension rather than two independent dimensions; i.e. the model captures the view that being 'excited' or 'enthusiastic' is incompatible with being 'hostile' or 'upset'.

A further model was tested (Model 1b) in which again all items were presumed to load upon only one general factor. However, PANAS items were drawn from Zevon and Tellegen's (1982) mood checklist, in which items are grouped into various categories based on content. In Model 1b, items from the same content categories were permitted to covary. These content categories are presented in brackets after the items they represent for PA: *attentive*, *interested* and *alert* (attentive); *enthusiastic*, *excited* and *inspired* (excited); *proud* and *determined* (proud); and *strong* and *active* (strong), and for NA: *distressed*, *upset* (distressed); *hostile*, *irritable* (angry); *scared*, *afraid* (fearful); *ashamed*, *guilty* (guilty); and *nervous*, *jittery* (jittery).

Models 2a–2e expressed variants on the hypothesis that the PANAS measures two factors, NA and PA. Model 2a represented the test authors' original conception of the dimensionality of the PANAS in that the ten PA items were indicators of a PA factor and the ten NA items were indicators of a NA factor. These two factors were constrained to be orthogonal, reflecting the original hypothesis (Watson & Clark, 1997) that 'variations in positive and negative mood are largely independent of one another' (p.270). Model 2b was identical to 2a except that the factors were allowed to correlate. This model reflects prior empirical evidence that NA and PA are moderately negatively correlated but posit that the overlap is not complete. Models 2c and 2d were identical to 2a and 2b, respectively, except that correlated error was permitted in accordance with the content categories from Zevon and Tellegen's (1982) mood checklist. For Model 2e, Model 2d was re-tested, but with '*excited*' permitted to cross-load upon NA as well as PA, as Mackinnon *et al.* (1999) found that making this adjustment led to a significant improvement in fit.

[1] *A computer program for PCs (sbdiff.exe.) that carries out this test is available. It can be downloaded from the following web page: http://www.abdn.ac.uk/~psy086/dept/sbdiff.htm*

Model 3a represented the hierarchical model reported by Mehrabian (1998). NA was conceptualized as a second-order factor consisting of two distinct first-order factors, *afraid* and *upset*. The former comprised six items (scared, nervous, afraid, guilty, ashamed, jittery) and the latter comprised four (distressed, irritable, hostile, upset). Technical difficulties were encountered when attempting to fit this model (empirical under-identification) that prevented a direct test of the hierarchical model. However, nested factor models were constructed, which although not mathematically equivalent, can be interpreted analogously. Instead of specifying higher order factors, first-order factors were constructed which possessed varying degrees of generality. Thus, although all ten indicators of subjective distress loaded on a general NA factor, six also loaded on a more specific 'afraid' factor, and the remaining four on a specific 'upset' factor. Two nested models were tested, in each of which the two specific factors, upset and afraid, were allowed to interrelate. The models were identical except that Model 3a did not permit the NA and PA factors to correlate, whereas Model 3b did.

Results

Testing competing confirmatory factor analytic models of the PANAS

The fit statistics for the CFA models are presented in Table 1. It can be seen that the general factor model (Model 1a) had very poor fit; the model's χ^2, SRMR and RMSEA are large and the RCFI low. Allowing correlated error between items derived from the same content categories in Zevon and Tellegen's (1982) mood checklist resulted in an improvement, but Model 1b also had extremely poor fit.

Model 2a expressed the original hypothesis that the PANAS measures two independent factors, PA and NA. The fit of this model is also very poor. Similar results

Table 1. Fit indices for CFA models of the PANAS (best fitting model in bold)

Model	S-Bχ^2	χ^2	*df*	RCFI	SRMR	RMSEA
1a. Single factor	2612.8	4102.7	170	0.55	.160	.152
1b. Single factor, correlated errors (CE) permitted	1298.3	1863.2	157	0.79	.140	.104
2a. Positive affect (PA) and negative effect (NA) as independent factors	1132.1	1589.9	170	0.82	.103	.091
2b. PA and NA as correlated factors	1089.9	1531.0	169	0.83	.066	.090
2c. PA and NA as independent factors, CE permitted	556.0	754.2	157	0.93	.096	.062
2d. PA and NA as correlated factors, CE permitted	**508.3**	**689.8**	**156**	**0.94**	**.052**	**.058**
2e. PA and NA as correlated factors, CE permitted; '*excited*' cross-loading	491.9	670.2	155	0.94	.050	.058
3a. Mehrabian's nested factors, NA and PA as independent factors	738.1	986.2	159	0.89	.097	.072
3b. Mehrabian's nested factors, NA and PA as correlated factors	685.3	932.1	158	0.90	.052	.070

Note. The Satorra-Bentler scaled χ^2 statistic (S-Bχ^2) was used to evaluate model fit. However, the normal χ^2 is also required when testing for a difference between the S-Bχ^2 statistic obtained from nested models; hence we present both statistics in this Table.

were obtained for Model 2b, in which PA and NA were permitted to correlate. Models 2c and 2d were counterparts of Models 2a and 2b, respectively, and differed only in that the models were parameterized to allow for the association (i.e. correlated error) predicted by Zevon and Tellegen's (1982) mood checklist.

Model permitting these correlated errors possessed markedly superior fit compared with their more constrained counterparts. As noted, inferential statistics can be applied in order to compare nested models. Models 2a and 2b are nested within 2c and 2d, respectively, in that they differ only by the imposition of the constraint that correlated errors are not permitted. The results from χ^2 difference tests used to compare these nested models are presented in Table 2.

Table 2. Results of testing for differences between nested CFA models of the PANAS

Comparison		Δ statistics		
More constrained	Less constrained	Δ S-Bχ^2	*df*	*p*
Model 1a	Model 1b	699.4	13	<.001
Model 2a	Model 2b	43.7	1	<.001
Model 2c	Model 2d	51.0	1	<.001
Model 2a	Model 2c	421.4	13	<.001
Model 2b	Model 2d	425.6	13	<.001

It can be seen that the models allowing correlated error had a significantly better fit ($p < .001$) than their more constrained counterparts. The fit of the correlated factors models (Models 2b and 2d) are also significantly better than their independent factor counterparts (Models 2a and 2c, respectively). Thus, the dimensions of PA and NA are, contrary to the test author's intentions, at least moderately interdependent ($r = -.30$, $p < .001$). It should be noted that the correlation between the NA and PA *factors* is higher than the correlation of $-.24$ ($p < .001$) between the NA and PA *scales*. This is because the NA and PA factors in the CFA models are estimated without error, whereas the correlation between the scales is attenuated by measurement error and the unique variance associated with each item.

Although it may initially appear that the general factor models are very different from the correlated factors models, they are also nested within these models. Models 2b and 2d can be rendered equivalent to their single-factor model counterparts simply by constraining the correlation between factors to unity (i.e. $r = 1.0$). The χ^2 difference tests comparing Model 1a with 2b and 1b with 2d were both highly significant, thereby demonstrating that it is untenable to view the PANAS as measuring only a single dimension.

Model 2d was associated with the optimal fit. The RCFI of .94 falls just short of Hu and Bentler's (1999) criterion, whilst the RMSEA (.058) and SRMR (.052) indicate a good fit. Moreover, Model 2d had a χ^2 value that, although statistically significant, was relatively small. (When dealing with large sample sizes and a moderate number of items, Byrne (1994) has pointed out that it is unusual to obtain non-significant χ^2 values for CFA models of self-report data.)

Model 2e was identical to Model 2d, but additionally permitted '*excited*' to cross-load upon NA. As in Mackinnon *et al.*'s (1999) study, this loading was not large (the loadings were .24 and .14, respectively), and thus this cross-loading was not retained in the optimal model. Moreover, it is important to note that this cross-correlation could not

account for the correlation between PA and NA; in Model 2e the correlation between these factors was the same as in Model 2d (−.30)

A schematic representation of the standardized solution for Model 2d is presented as Fig. 1. By convention, latent factors are represented by large ovals or circles, errors as smaller ovals or circles (as they are also latent variables) and manifest (i.e. observed) variables as rectangles or squares. Single-headed arrows connecting the variables represent a causal path, double-headed arrows represent covariance or correlation between variables, but do not imply causality.

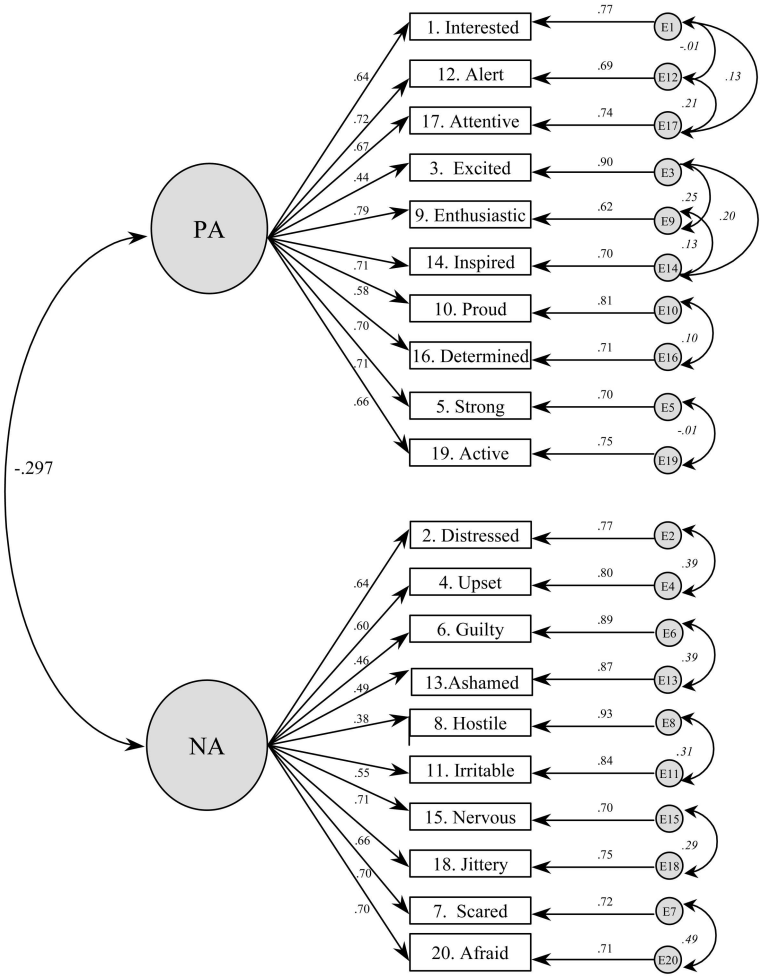


Figure 1. Graphical representation of the correlated two-factor model of the PANAS (Model 2d); the factor loadings are standardized loadings.

It should be noted that there are some authorities on structural equation modelling that consider permitting correlated error terms for subgroups of items from the same measurement instrument to be inappropriate. However, we considered that, for the PANAS, these correlated errors were appropriate because (a) they were specified *a priori* on the basis of theory and prior empirical findings, (b) with 20 items there are 180 potential correlated errors yet we only permit 13, thus the model is far from being

fully saturated, and (c) it can be seen that the correlations between items, especially on the NA scale, are moderate in magnitude (see Fig. 1). Thus, we have chosen to retain these correlated errors. Moreover, when we randomly split the sample into two, for both subgroups Model 2d was associated with the optimal fit. The associated RCFIs, SRMR and RMSEA ranged from .932 to .937, .053 to .060 and .059 to .063, respectively. It is also important to note that the introduction of correlated residuals did not substantially alter the values of factor loadings or the correlation between NA and PA.

Mehrabian's (1998) three-factor model, in both its independent factors (Model 3a) and correlated factors form (Model 3b), had a lower RCFI and a higher χ^2 and RMSEA than Model 2d did. The problems with these models were particularly apparent when the item loadings were examined. In Model 3b, for example, 12 of the factor loadings were below .5, all but one of which concerned items derived from the NA scale. Moreover, 6 of the 10 items loaded higher on the NA dimension than the relevant second-order factor.

Influence of demographic variables on PANAS scores and measurement invariance

Independent samples *t*-tests revealed that females obtained significantly higher scores than males on the NA scale (t (1,001) = 4.02, $p < .001$), but that males obtained significantly higher scores than females on the PA scale (t (1,001) = 3.00, $p = .003$).

To examine the influence of the remaining demographic variables on the PA scale, three hierarchical regression analyses were performed for each of the demographic predictor variables (age, years of education and occupational code). In each of these, the demographic variable (e.g. age) was entered into the regression model with subsequent entry of polynomial functions of the relevant predictor variable (e.g. age^2 followed by age^3). These analyses were then repeated for the NA scale. For all three demographic variables, on both PANAS scales, polynomial functions did not significantly increase the variance predicted. Therefore, there is no evidence of non-linear components in the relationships between demographic variables and PANAS scores, and these relationships can validly be expressed as correlation coefficients. The correlations between demographic variables and PANAS scores are presented in Table 3. The point–biserial correlation between gender and PANAS scores are also presented in this table as an index of effect size (males were coded as 0, females as 1; therefore, a positive correlation represents higher scores in females).

Table 3. Correlations between demographic variables and PANAS scores

	PANAS Scale	
Demographic variable	PA	NA
Age	.05	−.15*
Occupational code	−.11*	−.05
Years of education	.09*	.08
Gender	−.09*	.13*

*Significant at the .01 level (two-tailed).

It is possible that interactions between the demographic variables would explain variance in PANAS scores. To investigate this, hierarchical regression was performed in which the four demographic variables were entered as a first block followed by the six

variables that coded their two-way interactions (to code interactions the variables are multiplied by each other). The change in R^2 was not significant when these latter variables were added ($F = 1.90$, $p = 0.08$ for PA; $F = 0.64$, $p = 0.70$ for NA).

The above analyses were conducted on the *observed* variables (i.e. PA and NA total scores). It is also important to examine whether instruments possess measurement and factorial invariance across demographic subgroupings (Byrne, 1989; Hoyle & Smith, 1994). This was tested by performing median splits to form three sets of two subgroups, namely: younger participants (<43 years, $N = 507$) versus older (≥43 years, $N = 507$); low (<14 years, $N = 507$) versus high (≥14 years, $N = 489$) years of education; and female ($N = 537$) versus male ($N = 466$). Simultaneous (i.e. multi-group) confirmatory factor analyses were then performed to test whether releasing equality constraints on parameters across subgroups led to a significant improvement in model fit (the standard maximum likelihood method was used for these analyses as the robust method for multi-group CFA has not yet been implemented in EQS).

Preliminary analyses revealed that the optimal model (Model 2d) had the best fit in each of the three sets of two subgroups. Byrne (1989) notes that testing for equivalence of error covariances is considered to be overly stringent and therefore analysis was restricted to testing whether the factor loadings were equivalent across groups (i.e. testing for measurement invariance) followed by testing for factorial invariance (i.e. testing whether the covariance between PA and NA was equivalent across groups). The results of testing for measurement invariance revealed that the PANAS possessed full measurement invariance across the two age groups but that for education the loading for item 5 ('strong') on the PA factor was not invariant, i.e. releasing the equality constraint on this loading improved model fit ($\chi^2 = 5.06$, $p = .024$; nor was item 13 ('ashamed') invariant across gender ($\alpha^2 = 4.65$, $p = .031$).

These results suggest that the PANAS possesses what Byrne (1989, 1994) has termed *partial* measurement invariance. However, only one equality constraint in each of these two analyses was significant and, if a Bonferroni correction were applied to the p values for these items (to control for the fact that three separate multi-group analyses had been run), neither would remain significant. Therefore, to all intents and purposes, the PANAS can be regarded as possessing measurement invariance.

To test for factorial invariance, the models were re-run with the removal of the equality constraints on item 5 in the age analysis and item 13 in the gender analysis. Imposing equality constraints on the covariance between PA and NA did not lead to a significant deterioration in model fit for the age or education analyses. However, this equality constraint did lead to a deterioration in fit for gender ($\chi^2 = 5.95$, $p = .015$); this result would still be significant after a Bonferroni correction, although only marginally. The covariance between PA and NA was significantly greater for males than for females (r between PA and NA = $-.31$ for males and $-.24$ for females).

Summary statistics and normative data for the PANAS

The means, medians, *SD*s and ranges for the PA and NA scales are presented in Table 4 for the total sample, and for males and females and females separately.

A Kolmogorov-Smirnov test confirmed the impression from visual inspection of the distribution of scores that the NA scale was not normally distributed (because of high positive skew); $Z = 5.196$, $p < .001$). The PA scale also had a slight positive skew; the Kolmogorov-Smirnov test just failed to attain significance at the 5% level ($Z = 1.72$, $p = .05$). Given the positive skew, particularly for the NA scale, use of the raw score

Table 4. Summary statistics for the PANAS for the total sample and males and females separately

	Median	Mean	*SD*	Range
Total Sample (*N* = 1,003)				
PA	32	31.31	7.65	10–50
NA	14	16.00	5.90	10–42
Females (*N* = 537)				
PA	31	30.62	7.89	10–50
NA	15	16.68	6.37	10–42
Males (*N* = 466)				
PA	32	32.06	7.31	10–48
NA	14	15.20	5.23	10–42

means and *SD*s from a normative sample cannot be used to estimate the rarity of an individual's score. Therefore Table 5 was constructed for conversion of raw scores on the PA and NA scales to percentiles.

Reliabilities of the PANAS

The reliabilities (internal consistencies) of the PANAS PA and NA scales were estimated using Crobach's α. Cronbach's α was .89 (95% *CI* = .88–.90) for the PA scale, and .85 (95% *CI* = .84–.87) for the NA scale.

Relationship with measures of anxiety and depression

The correlations of the PA and NA scales with each of the anxiety and depression scales are presented in Table 6 and reveal that, as predicted, PA is more strongly negatively related to depression than to anxiety for both the DASS and the HADS.

However, all correlations in Table 6 are highly significant ($p < .01$), primarily as a consequence of the large sample size conferring high statistical power. Therefore, Williams' (1959) test was used to enable quantitative comparisons between correlations. This revealed that for both measures the correlation between PA and depression was significantly higher than that between PA and anxiety (DASS: t (986) = 7.523, $p < 001$; HADS: t (737) = 7.667, $p < .001$).

Two hierarchical regression analyses were also performed to examine the extent to which depression variance in the DASS could be explained by PA and NA. In order to examine the ability of the measures to explain variance *unique* to depression, shared variance was partialled out by entering DASS anxiety scores as the first predictors in the regression models. This was then followed by entry of either PA (Model A) or NA (Model B). The results are presented in Table 7, and indicate that both PA and NA were significant predictors of variance unique to depression. However, PA accounted for nearly twice as much unique variance as NA (8.3% compared with 4.7%). Steiger's (1980) test revealed that the proportion explained by PA was significantly greater than that explained by NA ($z = 2.20$, $p = .028$). This procedure was then repeated for the HADS, the results of which are also presented in Table 7 (Model C corresponds to entry of PA, Model D to entry of NA). Again, both PA and NA were significant predictors of variance unique to depression. However, the difference in proportion of variance accounted for was even more substantial than for the DASS; PA accounted for 14.7%,

Table 5. Raw scores on the PANAS converted to percentiles

Raw score	PA percentile	NA percentile
10	1	12
11	1	18
12	1	28
13	1	38
14	2	47
15	2	55
16	3	63
17	3	69
18	5	74
19	7	78
20	8	81
21	10	84
22	13	86
23	15	88
24	18	90
25	21	91
26	24	92
27	28	93
28	32	94
29	36	95
30	41	96
31	46	97
32	52	97
33	57	98
34	62	98
35	67	99
36	72	>99
37	77	>99
38	81	>99
39	85	>99
40	88	>99
41	90	>99
42	92	>99
43	94	>99
44	95	>99
45	97	>99
46	98	>99
47	99	>99
48	>99	>99
49	>99	>99
50	>99	>99

Note. In most clinical contexts, the concern will be whether NA scores are unusually high (i.e. the percentile is towards the top end of the scale) and whether PA scores are unusually *low* (i.e. the percentile is towards the lower end of the scale)

Table 6. Correlations between PA and NA with depression, anxiety and stress

Measure	DASS depression	HADS depression	DASS anxiety	HADS anxiety	DASS stress
PA	−.48	−.52	−.30	−.31	−.31
NA	.60	.44	.60	.65	.67

Note. All correlations significant at the .01 level (two-tailed).

Table 7. Hierarchical multiple regressions to examine the extent to which variance in depression scores can be explained by PA and NA

Criterion	Predictor	*R*	R^2 change	*p* value for R^2 change
Model A (DASS depression)				
Step 1	DASS anxiety	.706	.499	<.001
Step 2	PA	.763	.083	<.001
Model B (DASS depression)				
Step 1	DASS anxiety	.706	.499	<.001
Step 2	NA	.739	.047	<.001
Model C (HADS depression)				
Step 1	HADS anxiety	.500	.250	<.001
Step 2	PA	.630	.147	<.001
Model D (HADS depression)				
Step 1	HADS anxiety	.500	.250	<.001
Step 2	NA	.523	.024	<.001

NA only 2.4%. This difference was highly significant ($z = 6.26$, $p < .001$) using Steiger's (1980) test.

Discussion

Competing models of the structure of the PANAS

The use of CFA to test competing models of the latent structure of the PANAS yielded results that help resolve some of the inconsistencies in the previous EFA and CFA literature. From the fit statistics in Table 1, it is clear that the hypothesis that the PANAS measures a single factor (Models 1a and 1b) is untenable. Model 2a, an orthogonal two-factor model, represented a poor, though significantly better, fit than either of the two foregoing models. This model encapsulated the test authors' original hypothesis that the PANAS measures two independent factors, PA and NA. However, allowing PA and NA to covary (Model 2b) significantly improved the fit of the model, revealing that PA and NA as measured by the PANAS are at least moderately interdependent.

Model 2c was identical to Model 2a, but additionally permitted correlated error in accordance with Zevon and Tellegen's (1982) mood checklist. This was associated with

a significantly better fit than Model 2a. However, Model 2d, which specified correlated NA and PA factors and allowed correlated error, represented the optimal fit, and a significantly better fit than Model 2c ($p < .001$). Two versions of Mehrabian's (1998) model were tested; one in which PA and NA were restricted to orthogonality (Model 3a), and one in which they were permitted to correlate (Model 3b). Both solutions were poorer fits than Model 2d with respect to the RCFI, RMSEA and χ^2. Moreover, the pattern of factor loadings suggested that the hierarchical structures specified were inappropriate.

The conclusion from the CFA modelling is therefore that the PANAS NA and PA scales index two distinct, but moderately negatively correlated, factors. Opinions are liable to vary with regard to the extent to which the presence of a negative correlation between the PA and NA factors compromises the validity of the PANAS scale and/or the validity of the constructs it was designed to assess. Watson *et al.* (1988b) have argued that the failure to find independent factors of PA and NA in previous research should be attributed to the inadequacies of the instruments employed, rather than because the underlying theoretical constructs are not orthogonal. Great care was taken in the selection of items for the PANAS in an attempt to devise a scale that would measure these putatively independent constructs. As the present results demonstrate that PA and NA are negatively correlated even when using an instrument specifically developed to yield independent scales, this seriously questions whether these constructs are in fact independent.

However, the present results do indicate that PA and NA are *relatively* independent and this is consistent with the results of EFA and CFA analyses that found PA and NA to be largely independent dimensions (Tellegen *et al.*, 1999). In the present study, the percentage of shared variance between the PA and NA latent factors is only 9.0% (i.e. $-.30^2$); the percentage of shared variance between the observed scales (5.8%) is even more modest. Some may even regard the relatively modest amount of shared variation between these factors as partly vindicating Watson *et al.*'s (1988b) position, given that intuition would suggest that these factors should be highly negatively correlated (Costa & McCrae, 1980). Feeling 'enthusiastic' and 'inspired', for instance, should be incompatible with simultaneously feeling 'upset' or 'distressed'.

Finally, as noted previously, Tellegen *et al.* (1999) found a structure that is very consistent with the one identified in the present article, but additionally included a third-order dimension of happiness-versus-unhappiness. We did not incorporate this general factor in our own structure because *none* of the items found to load highly on the positive and negative poles of the third-order happiness-versus-unhappiness dimension (*at ease*, *happy* and *joyful* for happy; *blameworthy*, *discouraged*, *sad* and *downhearted* for unhappy) are included in the standard version of the PANAs we employed. However, future research should attempt to assess whether these same markers of this construct emerge across different samples.

Influence of demographic variables

The results of hierarchical regression analyses demonstrated that significant non-linear components were absent in the relationships between demographic variables and the PANAS (and also that interactions between the demographic variables did not account for variance in PANAS scores). Moreover, although six out of the eight linear relationships were significant, the effect sizes were very modest. The variance explained ranged from 0.20% (age and PA) to 2.25% (age and NA). Thus, for practical

purposes, the influence of gender, occupation, education and age on PANAS scores can be ignored; the significant effects result from the high statistical power conferred by a large sample size. This simplifies clinical use of the PANAS scores, as these variables (and their interactions) do not need to be taken into consideration when interpreting an individual's scores.

Normative data

To date, the PANAS has been used primarily as a research took in group studies. However, the instrument has the potential to be useful in clinical work with individuals. Prior to obtaining the present normative data, interpretation of the PANAS in the individual case relied on use of the means and *SD*s from a predominantly student sample (Watson *et al.*, 1988b). The current study usefully complements this by providing estimates of the degree of rarity of a given PANAS score in the general adult population. However, although the present sample was large and broadly representative of the general adult population in terms of basic demographic characteristics, it was not obtained by random sampling (e.g. from the electoral roll or a related method) and may therefore be subject to unforseen biases.

In the present study, the 'past week' time-frame was adopted because it is the one most commonly used in clinical populations. To our knowledge, normative data derived from non-clinical populations have not previously been provided for this time-frame. However, Watson *et al.* (1988b) reported mean scores for the closest available comparison ('past few days') in their US student sample; the PA mean was 33.3 (*SD* = 7.2), the NA mean 17.4 (*SD* = 6.2). This is broadly consistent with the results in the present sample in which the mean for PA was 31.3 (*SD* = 7.7) and the mean for NA was 16.0 (*SD* = 5.9).

The tabulation method in Table 5 was adopted to permit conversion of raw scores to percentiles for both PANAS scales using the same table. Thus for example, if a patient' raw score on the NA scale was 30, then Table 5 reveals that this corresponds to the 96th percentile; i.e. a score as high as this is estimated to be rare in the general adult population. A raw score of 17 on the PA scale corresponds to the 3rd percentile; i.e. a score as *low* as this is also rare in the general adult population.

According to tripartite theory (e.g. Clark & Watson, 1991b), patients diagnosed as either anxious or depressed will all experience high NA. It is argued that what differentiates depression from anxiety is the additional presence of *low* PA (i.e. loss of interest, and loss of the ability to experience pleasure). The normative data presented here should assist the clinician in interpreting the PANAS by providing estimates of the degree of rarity or abnormality of a client's NA and PA scores.

We suggest that the PANAS can be used as a supplement to measures of anxiety and depression to examine the extent to which they provide convergent evidence. For example, if a client obtains a high anxiety score but low to moderate depression score, the PANAS scores can be examined to see whether the patient scores high on NA (e.g. $\geq$95th percentile) combined with a PA score in the normal range or above (e.g. $\geq$20th percentile). When a patient scores highly on both anxiety and depression, then the PANAS can be examined to determine whether this is accompanied by indications of abnormally high NA (e.g. $\geq$95th percentile) and abnormally *low* PA (e.g. $\leq$5th percentile).

The comparison of anxiety and depression scores with the PANAS may be best achieved by using the Depression Anxiety and Stress Scales (DASS; Lovibond &

Lovibond, 1995), Hospital Anxiety and Depression Scale (HADS; Snaith & Zigmond, 1994; Zigmond & Snaith, 1983) or the Symptoms of Anxiety and Depression scale (sAD; Bedford & Foulds, 1978). Normative data are now available for all three scales in a form that is equivalent to that presented here for the PANAS (Crawford & Henry, 2003; Crawford, Henry, Crombie, & Taylor, 2001; Henry, Crawford, Bedford, Crombie, & Taylor, 2002); that is, they consist of percentile tables derived from large samples of the general adult population. It is therefore possible for a clinician to directly compare the rarity of a patient's PANAS scores with the rarity of their scores on these anxiety and depression scales. Such a comparison is likely to be more useful than comparing the PANAS to categorical or dichotomous indices (such as whether or not a patient is above a particular cut-off score for identifying caseness) derived from these or other anxiety and depression scales.

Reliabilities

The reliabilities of the PANAS scales, as measured by Cronbach's α, were .89 for PA and .85 for NA. The narrowness of the confidence limits associated with these coefficients indicate that they can be regarded as providing very accurate estimates of the internal consistency of the PANAS in the general adult population. Thus, both PA and NA scales can be viewed as possessing adequate reliability.

Relationships with measures of anxiety and depression

Correlational analyses were conducted in order to test the prediction of tripartite theory that PA is more strongly related to depression than to anxiety (Watson, Clark, & Carey, 1988a). The results were supportive, revealing that although all correlations were significant, for both the DASS and the HADS the correlation between PA and depression was significantly higher than that between PA and anxiety. Regression analyses were then conducted to examine how much variance *unique* to depression was explained by NA and PA, respectively. For both the DASS and the HADS, although PA and NA each explained a significant proportion of variance unique to depression, PA explained significantly more. These results are therefore broadly consistent with the tripartite model, indicating that PA is a significantly better predictor of variance unique to depression than is NA. Moreover, the difference in the proportion of variance accounted for was substantially larger for the HADS than the DASS. This latter finding is probably attributable to the greater emphasis that the HADS places upon the state of anhedonia, as tripartite theory regards PA and anhedonia as highly (negatively) related (Clark & Watson, 1991a).

Conclusions and future research

To conclude, the PANAS has been shown to possess adequate psychometric properties in a large sample drawn from the general adult population. The results from CFA modelling largely support the construct validity of the PANAS scales and the reliabilities of both scales were adequate. The norms presented are, to our knowledge, the only UK norms currently available.

Although the inconsistencies in the previous literature on the factor structure of the PANAS could, in part, be attributed to differences in the methodology employed, it may

also be that the structure of the PANAS is not invariant across different populations. Hoyle and Smith (1994) and others (Byrne, 1989), have observed that in psychology we have often neglected the question of whether our instruments are factorially invariant. Instead, we have commonly simply assumed that they are measuring the same constructs across different populations. If, for a particular instrument, this assumption is incorrect, than comparing the scores of groups or individuals is 'a classic example of comparing apples and oranges' (Hoyle & Smith, 1994, p. 433).

The present results suggest that the PANAS may not possess factorial invariance across gender. However, we do not consider that this poses a serious threat to the validity or utility of the PANAS given that (1) the PANAS exhibited measurement invariance across gender, (2) the test on the equivalence of the factor covariances showed them to be only marginally significant after a Bonferroni correction, and (3) the magnitude of the difference in covariances/correlations between PA and NA was modest. However, it would be very valuable to examine whether the PANAS possesses measurement and factorial invariance across cultures and, more importantly, across healthy and clinical populations.

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